

STROMMINGSBADN, Finland

WMO: 027810

Lat: 62.979N

Lon: 20.740E

Elev: 4

StdP: 101.28

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-17.8	-13.7	-20.2	0.6	-17.9	-15.8	0.9	-13.8	17.9	-0.5	16.7	-1.2	5.0	50	0.862

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	2.6	21.8	19.7	20.0	18.0	18.7	16.8	19.8	21.8	18.4	19.8	17.3	18.5	4.9	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.9	13.7	21.2	17.7	12.7	19.6	16.7	11.9	18.2	56.7	22.0	52.0	19.7	48.6	18.5	22.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 16.5	14.9	13.5	DB	-17.8	23.0	5.8	2.2	-22.0	24.6	-25.3	25.9	-28.6	27.1	-32.8	28.7	(4)
(5)			WB	-17.3	19.8	6.2	1.7	-21.7	21.1	-25.4	22.1	-28.8	23.1	-33.3	24.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	5.6	-3.2	-3.6	-1.3	2.0	7.1	11.5	15.7	15.9	12.5	6.9	3.2	0.0	(6)
(7)		DBStd	7.55	4.47	5.49	3.25	1.92	2.79	2.34	2.65	2.26	2.47	2.54	3.33	4.38	(7)
(8)		HDD10.0	2110	409	381	351	240	99	10	0	0	8	98	204	309	(8)
(9)		HDD18.3	4656	667	614	609	490	348	204	91	83	174	354	454	567	(9)
(10)		CDD10.0	511	0	0	0	0	9	57	176	182	84	3	0	0	(10)
(11)		CDD18.3	14	0	0	0	0	0	0	8	6	0	0	0	0	(11)
(12)		CDH23.3	13	0	0	0	0	0	0	8	6	0	0	0	0	(12)
(13)		CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	7.0	7.8	7.5	6.9	6.0	6.1	5.9	5.6	6.2	7.1	8.0	8.3	8.6	(14)
(15)	Precipitation	PrecAvg	754	65	50	48	40	47	62	78	80	67	71	74	71	(15)
(16)		PrecMax	973	115	96	89	79	110	101	150	134	141	162	141	124	(16)
(17)		PrecMin	591	10	6	12	13	10	23	14	26	25	11	24	19	(17)
(18)		PrecStd	100	25	24	19	19	22	20	34	33	28	32	29	26	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.2	3.7	4.1	8.6	16.0	19.1	24.5	23.7	18.0	13.1	9.9	7.0	(19)
(20)			MCWB	3.2	2.3	2.6	6.3	13.2	16.3	21.0	20.8	16.5	11.3	8.6	6.0	(20)
(21)		2%	DB	3.4	2.9	3.3	6.7	14.0	17.3	22.3	21.9	16.6	12.1	9.0	6.1	(21)
(22)			MCWB	2.5	1.8	2.0	5.2	11.5	15.4	20.1	19.7	15.0	10.6	8.2	5.0	(22)
(23)		5%	DB	2.7	2.1	2.8	5.3	12.3	15.8	20.7	19.9	16.1	10.8	8.0	5.2	(23)
(24)			MCWB	1.8	1.3	1.7	4.0	10.3	14.0	18.9	17.4	14.6	9.5	7.5	4.0	(24)
(25)		10%	DB	1.8	1.6	2.2	4.4	10.9	14.6	19.2	18.6	15.6	10.1	7.1	4.4	(25)
(26)			MCWB	0.9	0.9	1.3	3.2	9.1	13.0	17.5	16.6	14.1	8.9	6.2	3.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.7	2.6	2.9	6.5	13.5	16.6	21.5	21.6	16.4	11.9	8.8	6.3	(27)
(28)			MADB	4.0	3.2	3.6	7.5	15.8	18.7	23.2	23.4	17.5	12.6	9.4	6.8	(28)
(29)		2%	WB	2.6	1.8	2.3	4.8	11.6	15.6	20.4	19.4	15.6	10.9	8.2	5.2	(29)
(30)			MADB	3.1	2.4	2.9	5.9	13.5	17.1	22.2	21.5	16.4	11.8	8.8	6.0	(30)
(31)		5%	WB	1.8	1.3	1.9	3.9	10.4	14.5	19.1	18.0	15.0	9.8	7.6	4.2	(31)
(32)			MADB	2.7	1.8	2.5	4.7	12.2	15.6	20.8	19.4	15.9	10.6	8.2	5.0	(32)
(33)		10%	WB	1.1	0.9	1.5	3.1	9.3	13.5	17.9	17.0	14.4	9.2	6.6	3.5	(33)
(34)			MADB	1.7	1.4	2.1	3.8	10.7	14.5	19.0	18.1	15.3	10.0	7.0	4.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	3.5	3.5	3.1	2.6	3.1	3.3	3.1	2.6	2.3	2.2	2.4	2.8	(35)
(36)			MCDBR	3.0	2.3	2.6	3.6	4.9	4.9	4.5	3.3	2.6	1.9	2.2	2.5	(36)
(37)			MCWBR	2.6	2.0	2.0	2.4	3.2	3.1	2.9	2.4	2.1	2.2	2.2	2.6	(37)
(38)		5% WB	MCDBR	2.7	2.1	2.4	3.2	4.9	4.3	4.1	3.0	2.4	1.8	2.1	2.5	(38)
(39)			MCWBR	2.5	2.1	1.9	2.3	3.4	3.0	2.9	2.6	2.1	2.2	2.0	2.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.243	0.297	0.317	0.356	0.355	0.352	0.374	0.362	0.336	0.319	0.384	0.349	(40)	
(41)		taud	2.075	2.192	2.265	2.319	2.447	2.498	2.462	2.484	2.536	2.391	2.678	2.554	(41)	
(42)		Ebn at Noon	389	634	780	824	862	872	838	816	772	618	360	218	(42)	
(43)		Edh at Noon	31	63	86	101	97	95	96	86	68	53	29	18	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.14	0.71	2.22	3.83	5.33	5.98	5.45	4.17	2.45	0.94	0.23	0.08	(44)	
(45)		RadStd	0.02	0.14	0.26	0.39	0.49	0.42	0.47	0.36	0.23	0.18	0.04	0.01	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(47) Regional (40 neighbors)	+0.42	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Nomenclature: See separate page